



Solutions Guide: Sysadmin Bash Script

To complete this activity, you needed to do the following:

- Add all the commands in the instructions to your script.
- Change the permissions on your script to make it executable.
- Run your script to verify it produces the correct output.

Solutions

Create a new script file.

- `touch sys_info.sh`

Change the permissions on the file to make it executable.

- `chmod +x sys_info.sh`

Open the file with nano.

- `nano sys_info.sh`

Add a top hashbang line to make this a bash script.

- `#!/bin/bash`

At this point your terminal output should look like:

```
touch sys_info.sh
```

```
chmod +x sys_info.sh
```

```
nano sys_info.sh
```

Then inside nano, you should have:

```
#!/bin/bash
```

Add the following to your script:

A title.

- `echo "A Quick System Audit Script"`

Today's date.

- `date`

The machine's type.

- `echo "Machine Type Info:"`
-
-

- `echo` expands any input it's given before sending it to the output.
- `$MACHTYPE` is a 'built-in' variable that contains the type of machine you are working on.

The uname info for the machine.

- `echo -e "Uname info: $(uname -a) \n"`
- `echo` sends everything to output.
- `-e` enables `echo` to read added line breaks within the line to be echoed.
- "Uname info: is printed out as shown.
- `$(uname -a)` is run before any other part of the line is run. This part gets run first. Then, its output is added to the line and the rest of the `echo` command is run.
- `\n` closes out the `echo` command and adds a line break. Note: this only works because we are adding the `-e` flag to `echo`.

The machine's IP address.

- `echo -e "IP Info: $(ip addr show eth 0 | head -n 3 | tail -n 1) \n"`

Let's breakdown this line:

`ip addr show eth0` is expanded.

- Run `ip addr show eth0`.

Your output should look similar to:

```
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
```

```
link/ether 00:16:3e:5e:6c:00 brd ff:ff:ff:ff:ff:ff
```

```
inet 10.137.0.21/32 brd 10.255.255.255 scope global eth0
```

```
valid_lft forever preferred_lft forever
```

```
inet6 fe80::216:3eff:fe5e:6c00/64 scope link
```

```
valid_lft forever preferred_lft forever
```

We want to narrow this output down to the line that contains our main IP address. In this case, it is the third.

- Run `ip addr | head -n 3`.

Your output should be:

```
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
```

```
link/ether 00:16:3e:5e:6c:00 brd ff:ff:ff:ff:ff:ff
```

- `inet 10.137.0.21/32 brd 10.255.255.255 scope global eth0`

Use `tail` to get the last line.

- Run `ip addr | head -n 3 | tail -n 1`.

Your output should be:

- `inet 10.137.0.21/32 brd 10.255.255.255 scope global eth0`

Now, surround this command with our expansion syntax `$()` so that it runs before `echo`:

- Run `echo -e "IP Info: $(ip addr | head -9 | tail -1) \n"`.

Your output should be similar to:

```
IP Info:  inet 10.137.0.21/32 brd 10.255.255.255 scope global eth0
```

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Let's return to the script.

The Hostname.

- `echo "Hostname: $(hostname -s) "`
 - The `-s` flag for `hostname` provides a 'short' hostname and is not absolutely required.

The final script should be similar to:

```
#!/bin/bash
```

```
echo "A Quick System Audit Script"
```

```
date
```

```
echo ""
```

```
echo "Machine Type Info:"
```

```
echo $MACHTYPE
```

```
echo -e "Uname info: $(uname -a) \n"
```

```
echo -e "IP Info: $(ip addr | grep inet | tail -2 | head -1) \n"
```

```
echo "Hostname: $(hostname -s) "
```

Bonuses:

The DNS info.

- `echo "DNS Servers: "`
- `cat /etc/resolv.conf`

The DNS info is stored in the `/etc/resolv.conf` file. All we need to do is display the contents of this file using `cat`.

The Memory info.

- `echo "Memory Info:"`
- `free`

The CPU info.

- `echo -e "\nCPU Info:"`
 - `echo -e "\nCPU Info:"` gives us a title with a line break before it.
- `lscpu | grep CPU`
 - `lscpu` gives us a ton of info about the computer's CPU.

- Remember, `ls` has a number of extended commands to show hardware and other system info.
- `| grep` pipes that output into `grep` so we can parse just the info we want.
- CPU is given to `grep` to display lines that only contain CPU.

The Disk usage.

- `echo -e "\nDisk Usage:"`
- `df -H | head -2`
 - `df` retrieves the disk information.
 - `-H` displays the info in human readable format. This means it will display bytes in megabytes and gigabytes instead of bytes.
 - `| head`: Again, we are piping the command into the `head` command to limit output.
 - `2` limits the output of `head` to 2 lines.

The current logged-in users.

- `echo -e "\nWho is logged in: \n $(who -a) \n"`
 - `echo -e "\n` initiates our `echo` command and creates a line break.
 - `Who is logged in: \n` will be printed as shown with another line break.
 - `$(who -a)` runs the `who -a` command before the `echo` command.
 - `\n` provides another line break.

At this point, our script should look like:

```
#!/bin/bash
```

```
echo "A Quick System Audit Script"
```

```
date
```

```
echo ""
```

```
echo "Machine Type Info:"
```

```
echo $MACHTYPE
```

```
echo -e "Uname info: $(uname -a) \n"
```

```
echo -e "IP Info: $(ip addr show eth0 | head -n 3 | tail -n 1) \n"
```

```
echo "Hostname: $(hostname -s) "
```

```
echo "DNS Servers: "
```

```
cat /etc/resolv.conf
```

```
echo "Memory Info:"
```

```
free
```

```
echo -e "\nCPU Info:"
```

```
lscpu | grep CPU
```

```
echo -e "\nDisk Usage:"
```

```
df -H | head -2
```

```
echo -e "\nWho is logged in: \n $(who -a) \n"
```

Close and save your script file.

Run your script using `./` notation.

- `./sys_info.sh`